

KSHITIJ GUPTA

Himachal Pradesh, India

☎ 8679930799 ✉ guptakshitij266@gmail.com 🔗 LinkedIn 🌐 GitHub 🌐 kshitij.dev

Summary

Automation Engineer with experience building scalable web scraping and automation solutions for social media platforms including YouTube, X (Twitter), Instagram, and Telegram. Contributed to the development of Axio Sentinel, an AI-powered brand monitoring and online reputation management platform. Experienced in Python, REST APIs, data pipelines, web automation, and backend integrations. Previously worked on IoT systems involving Raspberry Pi, MQTT, Docker, ChirpStack, and Node-RED during research at IIT Mandi.

Education

University School of Studies, Himachal Pradesh Technical University <i>Bachelor of Technology in Computer Science & Engineering</i>	Jul. 2023 – Jul. 2026 <i>Himachal Pradesh, India</i>
Government Polytechnic Sundernagar <i>Diploma in Computer Science Engineering</i>	Aug. 2019 – Jun. 2022 <i>Himachal Pradesh, India</i>
B.L. Central Public School, Karsog (CBSE) <i>Secondary School (Class X)</i>	May 2018 – May 2019 <i>Himachal Pradesh, India</i>

Experience

Automation Engineer <i>Axio Principle</i>	May 2026 – Present <i>Gurugram, India</i>
– Developed and maintained automated web scraping solutions for multiple social media platforms including YouTube, X (Twitter), and Instagram. – Built data collection pipelines to extract public posts, videos, channels, profiles, engagement metrics, and metadata for large-scale monitoring and analysis. – Contributed to the development of Axio Sentinel, an AI-powered brand monitoring and online reputation management platform for detecting brand mentions, piracy, and harmful content. – Designed automation workflows for scheduled data collection, processing, and storage using Python, REST APIs, and database integrations. – Integrated scraped data with backend services to enable real-time monitoring, analytics dashboards, and reporting. – Optimized scraping performance by implementing request handling, retry mechanisms, rate limiting, and error recovery to improve reliability.	
Research Intern – IoT & Network Systems (LoRaWAN Deployment) <i>Indian Institute of Technology (IIT) Mandi</i>	Jan. 2026 – May 2026 <i>Himachal Pradesh, India</i>
– Installed FreeBSD and configured Jails for deploying ChirpStack-based LoRaWAN services. – Configured LoRaWAN gateways and end devices for long-range sensor data collection and monitoring. – Integrated MQTT and Raspberry Pi for real-time sensor data acquisition and communication. – Optimized sensor node configurations to reduce power consumption and improve battery life.	
Research Intern – IoT & Healthcare Systems <i>Indian Institute of Technology (IIT) Mandi</i>	Jun. 2025 – Aug. 2025 <i>Himachal Pradesh, India</i>
– Developed IoT-based non-invasive healthcare monitoring systems using Raspberry Pi and sensors. – Designed real-time data pipelines using Node-RED and MQTT. – Visualized data using Grafana with InfluxDB and deployed services via Docker containers.	
Technician – Computer Engineering <i>Directorate of Technical Education</i>	Sep. 2022 – Aug. 2023 <i>Sundernagar, Himachal Pradesh</i>
– Provided technical support for computer hardware, operating systems, and lab infrastructure. – Assisted in system troubleshooting, installations, and basic network configuration.	

Projects

School ERP System	2026
<i>Full-Stack Web Application</i>	<i>India</i>
– Developed a web-based School ERP System for admissions, attendance, fee management, examinations, and result processing. – Implemented role-based authentication, responsive dashboards, and REST APIs for administrators, teachers, students, and parents. – Live: https://www.usvtutorkataula.co.in/	
LoRaWAN Infrastructure Setup using ChirpStack and FreeBSD	Jan. 2026 – May 2026
<i>IoT & Network Systems Project</i>	<i>India</i>
– Configured ChirpStack on FreeBSD Jails with Raspberry Pi, MQTT, and LoRaWAN gateways. – Optimized communication parameters to improve network reliability and sensor battery life.	
Smart Inventory Management System	2025
<i>IoT & Computer Vision Project</i>	<i>India</i>
– Developed an OCR-based inventory management system using computer vision for real-time stock monitoring. – Integrated backend synchronization for automated inventory updates and reporting.	
IoT Bazaar	2025
<i>Android Full-Stack Application</i>	<i>India</i>
– Built an Android application using Kotlin and Jetpack Compose with Node.js and MongoDB backend. – Implemented authentication, product catalog, search, cart, and order management features.	
Smart Smoke Detection and Alert System	2025
<i>IoT Embedded Systems Project</i>	<i>India</i>
– Developed an IoT-based smoke detection system using NodeMCU, MQ-2, and SIM800L. – Implemented SMS alerts and real-time local warning using buzzer, LEDs, and LCD display.	
Mini CNC Portrait Drawing Machine	2024
<i>Embedded Systems Project</i>	<i>India</i>
– Built an Arduino Nano-based CNC plotter capable of converting images into automated sketches.	

Certifications

Cloud Computing	Jun. 2025
<i>IBM</i>	<i>Online</i>
Introduction to Containers, Kubernetes & OpenShift	Jun. 2025
<i>IBM</i>	<i>Online</i>
Prompt Engineering	Jun. 2025
<i>IBM</i>	<i>Online</i>
Android App Development	Jul. 2024
<i>Internshala Trainings</i>	<i>Online</i>
Android App Development	Oct. 2021
<i>Solitaire Infosys</i>	<i>India</i>

Leadership & Positions of Responsibility

President – Boys Hostel	2021 – 2022
<i>Government Polytechnic Sundernagar</i>	<i>Himachal Pradesh, India</i>
– Led hostel administration and coordinated between students and faculty. – Handled discipline, facilities, and student welfare activities.	

Extracurricular Activities

Participant – Virtual Labs Interaction Program	2022
<i>Indian Institute of Technology (IIT) Delhi</i>	<i>India</i>
– Gained exposure to simulation-based and remote laboratory platforms.	